

□ CHAPTER 5: Project Development

5.1 Python Ingestion Controller Framework (app.py)

```
import os
import random
from datetime import datetime
from flask import Flask, render_template, request

app = Flask(__name__)

MODELS = {
    "Decision Tree": {"accuracy": 0.82, "id": 101},
    "Random Forest": {"accuracy": 0.87, "id": 102},
    "KNN": {"accuracy": 0.79, "id": 103},
    "XGBoost": {"accuracy": 0.89, "id": 104}
}

def preprocess_and_predict(data, model_name):
    try:
        income = float(data.get("income", 0))
        loan_amount = float(data.get("loan_amount", 0))
        credit_history = int(data.get("credit_history", 1))
    except (ValueError, TypeError):
        return "Rejected", 0.05

    base_score = 0.4 if credit_history == 1 else 0.05
    accuracy_modifier = MODELS.get(model_name, {"accuracy": 0.8})["accuracy"]
    probability = min(max(base_score * accuracy_modifier + random.uniform(-0.03,
0.03), 0.01), 0.99)
    return ("Approved" if probability >= 0.50 else "Rejected"), round(probability,
4)
```